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ITWS 4600: Data Analytics

25 April 2026

Predictive and Prescriptive Data Analytics: Absenteeism at Work

1. Exploratory Data Analysis

1.1 Dataset Overview

The Absenteeism at Work dataset contains 740 records and 21 variables covering employee demographics, job characteristics, lifestyle factors, and absenteeism hours. There are no missing values in the dataset, so no imputation was required.

1.2 Summary Statistics

The table below summarizes key numeric variables after outlier removal.

Variable	Mean	Std Dev	Min	Max
Absenteeism (hours)	6.2	8.6	0	40
Age (years)	36.5	6.5	27	58
Transport Expense (\$)	221.3	67.0	118	388
Distance to Work (km)	29.6	14.8	5	52
Body Mass Index	26.7	4.3	19	38
Workload (avg/day)	271.5	39.1	205.9	378.9

1.3 Distribution and Outlier Removal

The target variable, absenteeism time in hours, is heavily right-skewed (mean = 6.9h, median = 3h, max = 120h). This indicates that most employees have moderate absences while a small number have very high absences. A $3\times\text{IQR}$ rule was applied to remove extreme outliers

(values above 40 hours), resulting in 712 usable records after removing 28 outliers. The Q-Q plot confirms that even after removal, the distribution retains a mild positive skew, which is expected for count-like time data.

Figure 1: Exploratory Data Analysis

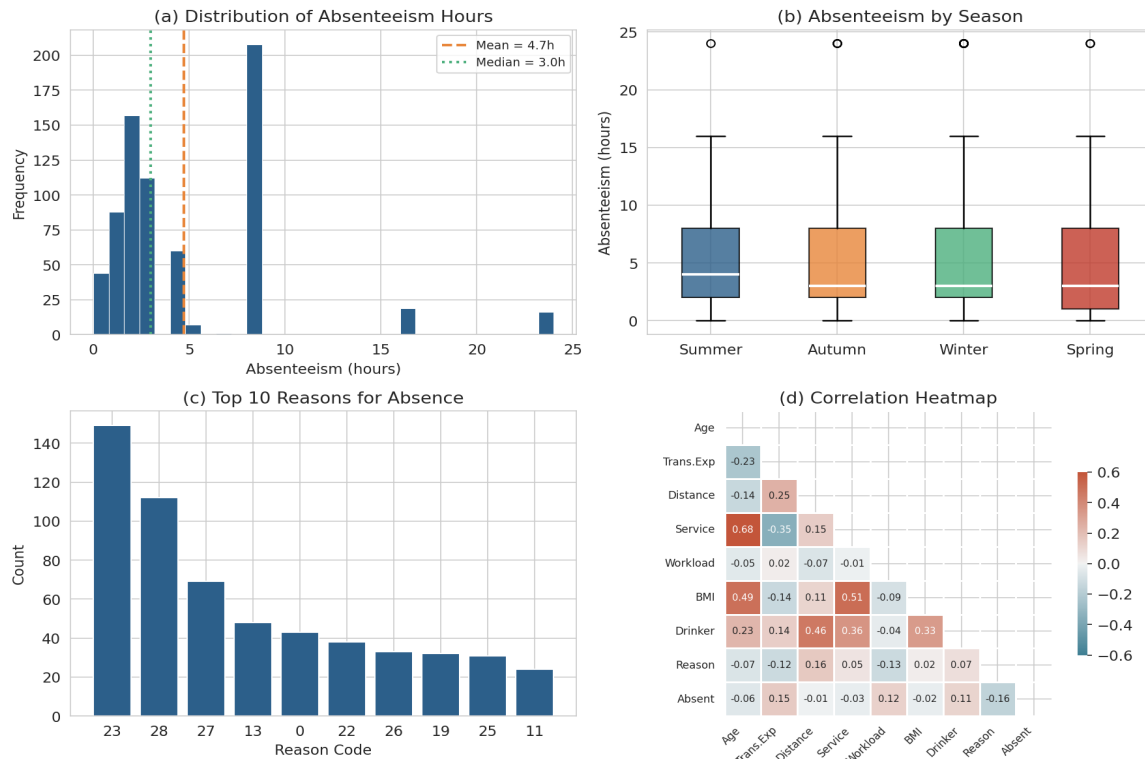


Fig. 1. EDA panels showing (a) distribution of absenteeism hours, (b) absenteeism by season, (c) top 10 absence reasons, and (d) correlation heatmap.

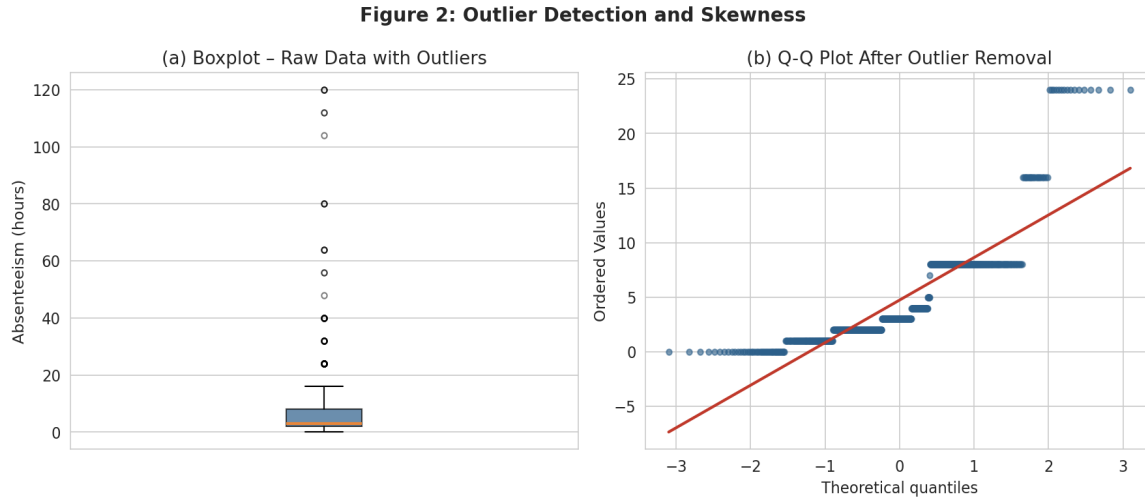


Fig. 2. (a) Raw boxplot showing extreme outliers; (b) Q-Q plot of cleaned data showing residual right skew.

1.4 Key Observations

Reason for absence (ICD categories) is the most frequent variable in absence records. Codes 23 (dental consultation) and 28 (physiotherapy) dominate, suggesting that medical appointments, not illness severity, drive most absences.

Seasonal variation is modest but visible; summer shows a slightly wider spread than other seasons, possibly driven by holiday-related absences.

Correlation analysis reveals that no single variable is strongly correlated with absenteeism hours (highest ≈ 0.30 for reason for absence). This means a non-linear ensemble model is likely a better fit than simple linear regression.

2. Model Development, Validation, and Optimization

Three models were built: a Random Forest Regression to predict the number of absenteeism hours, a Random Forest Classification to predict whether an employee falls in the high-absenteeism group, and K-Means Clustering to identify natural employee groupings. For each model, both the full feature set (17 variables) and a PCA-reduced feature set were tested to assess the impact of dimensionality reduction.

2.1 Feature Selection and Response Variables

The 17 input features include personal attributes (age, education, BMI), workplace factors (workload, service time, hit target), lifestyle indicators (social drinker/smoker, pets), and travel factors (distance, transportation expense). The reason for absence code was kept as a numeric feature since it was found to be the most important predictor.

Regression target: Absenteeism time in hours (continuous).

Classification target: High absenteeism (binary: 1 if hours $\geq$ 8h, i.e., top quartile, else 0). This threshold captures the 34.1% of employees who are disproportionately absent, making it a meaningful and data-driven boundary.

Clustering: No target variable; the goal is to find natural employee groups based on a subset of seven behaviorally meaningful variables.

2.2 Dimensionality Reduction via PCA

All features were standardised (zero mean, unit variance) before PCA. Ten principal components were required to explain 85% of the variance in the dataset, reducing the feature space from 17 to 10 dimensions. As shown in Figure 3b, the first two PCs do not cleanly separate

high and low absenteeism classes, which hints that the relationship is spread across many components and is inherently difficult to model with simple linear boundaries.

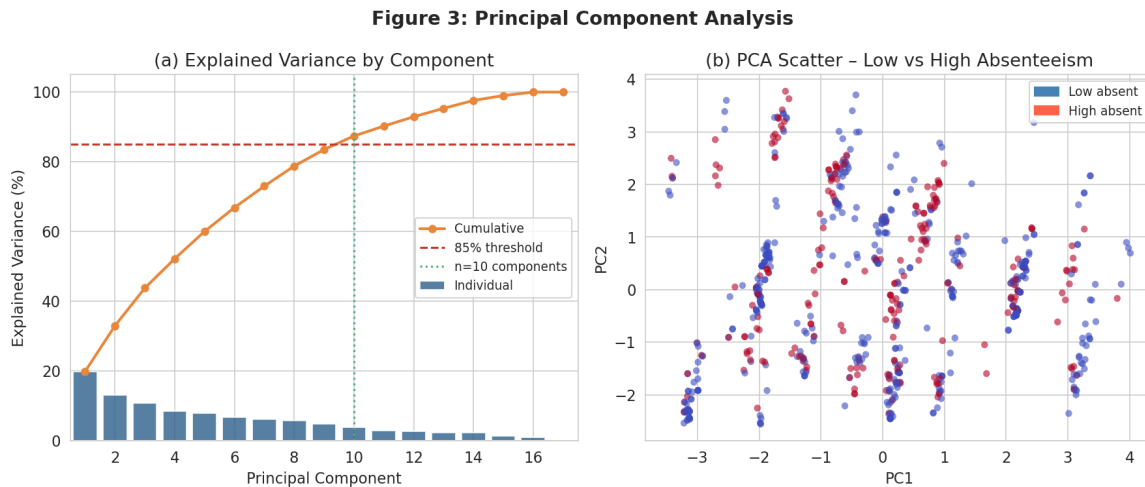


Fig. 3. (a) Scree plot showing cumulative explained variance; 10 PCs reach the 85% threshold.

(b) PCA scatter plot coloured by absenteeism class shows overlapping groups.

2.3 Validation Approach

All models were validated using an 80/20 train-test split and additionally evaluated with 5-fold cross-validation on the full dataset to assess generalisation. The train-test split gives a held-out performance estimate, while cross-validation reduces the risk of an overly optimistic result due to a lucky split.

2.4 Model 1 – Random Forest Regression

A Random Forest with 200 trees was trained to predict absenteeism hours. The model captures non-linear interactions and variable importance without requiring feature transformations.

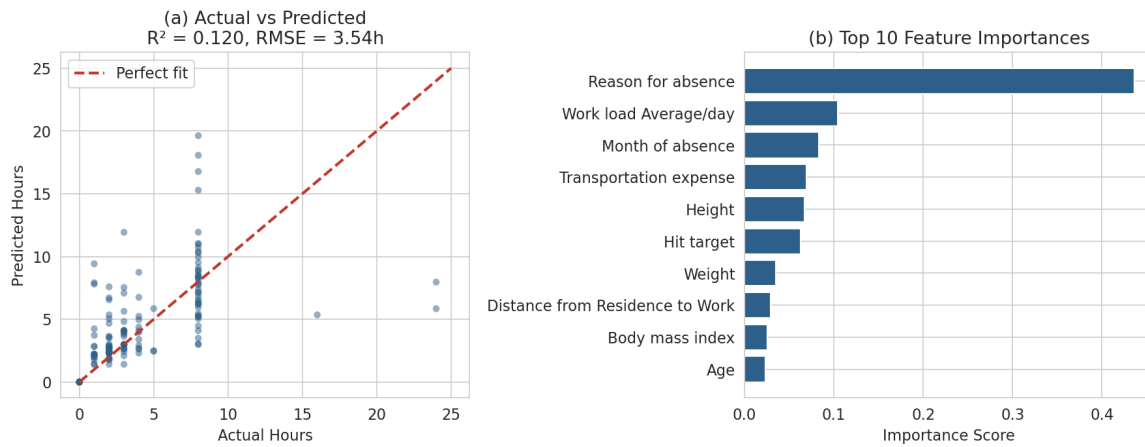
Figure 4: Random Forest Regression Results

Fig. 4. (a) Actual vs. predicted plot for Random Forest Regression; (b) top 10 feature importances showing reason for absence as dominant predictor.

The model achieved an R^2 of 0.120 on the test set, with a Root Mean Square Error of 3.54 hours. The 5-fold cross-validation R^2 was higher at 0.308, suggesting some variability between splits. The low R^2 on the test set indicates that raw hour prediction is difficult given the noisy and skewed nature of the target. The PCA version performed comparably with an R^2 of 0.282 and RMSE of 3.20 hours, suggesting that dimension reduction did not significantly hurt regression performance.

Metric	Full Features	PCA Features	5-Fold CV (Full)
R^2	0.120	0.282	0.308
RMSE (hours)	3.54	3.20	—

2.5 Model 2 – Random Forest Classification

A Random Forest Classifier with 200 trees and balanced class weights was trained to predict whether an employee falls in the high-absenteeism group (≥ 8 hours). Using the top quartile ($Q75 = 8h$) as the decision boundary, 34.1% of employees are labelled “High.” Class weights were set to handle this imbalance.

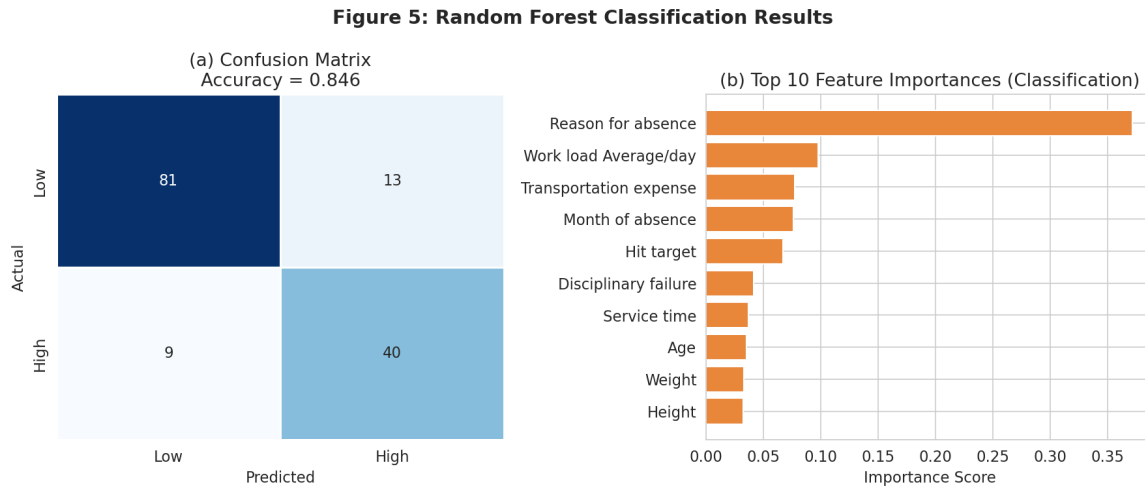


Fig. 5. (a) Confusion matrix showing 85% accuracy; (b) feature importances for the classification task.

The classifier achieved 84.6% accuracy on the test set and 77.7% in 5-fold cross-validation. The precision for the “High” class was 75% and recall 82%, indicating that the model catches most high-risk employees while producing some false positives. The PCA version achieved a slightly lower accuracy of 76.9%, showing a moderate cost to dimensionality reduction for this task.

Metric	Full	PCA	CV (Full)	Prec. High	Rec. High	F1 High
Accuracy	84.6%	76.9%	77.7%	75%	82%	78%

2.6 Model 3 – K-Means Clustering

K-Means clustering was applied to seven variables: reason for absence, transportation expense, age, workload, BMI, social drinker status, and absenteeism hours. Variables were standardized before clustering. The optimal number of clusters was selected using the silhouette score, which measures how well each point fits within its assigned cluster.



Fig. 6. (a) Elbow method; (b) silhouette scores indicating $K=8$ as optimal; (c) cluster assignments visualized in PCA space.

The silhouette analysis identified $K = 8$ as optimal (silhouette score = 0.250). Among the eight clusters, one stands out with an average absenteeism of 19.7 hours, nearly four times the overall mean. This cluster likely represents employees with chronic or serious medical conditions. Other clusters group employees with low absenteeism (around 2.8 to 3.4 hours) but differ in age, workload, and lifestyle factors.

Cluster	Reason	Trans. Exp.	Age	Workload	BMI	Avg Hours
0	24.1	200.6	36.0	260.6	27.2	3.4
1	17.3	224.1	37.1	279.8	26.3	5.4
2	20.8	213.8	35.2	271.2	26.8	4.4
3	4.3	261.7	35.1	270.2	26.8	4.3
4	22.2	192.2	38.0	265.1	26.1	2.9
5	23.5	241.6	37.2	276.6	26.5	5.0
6*	14.6	242.8	36.7	278.8	27.1	19.7
7	12.3	219.8	36.3	272.0	26.8	2.8

* Cluster 6 has notably high average absenteeism (19.7 hours).

2.7 Model Comparison

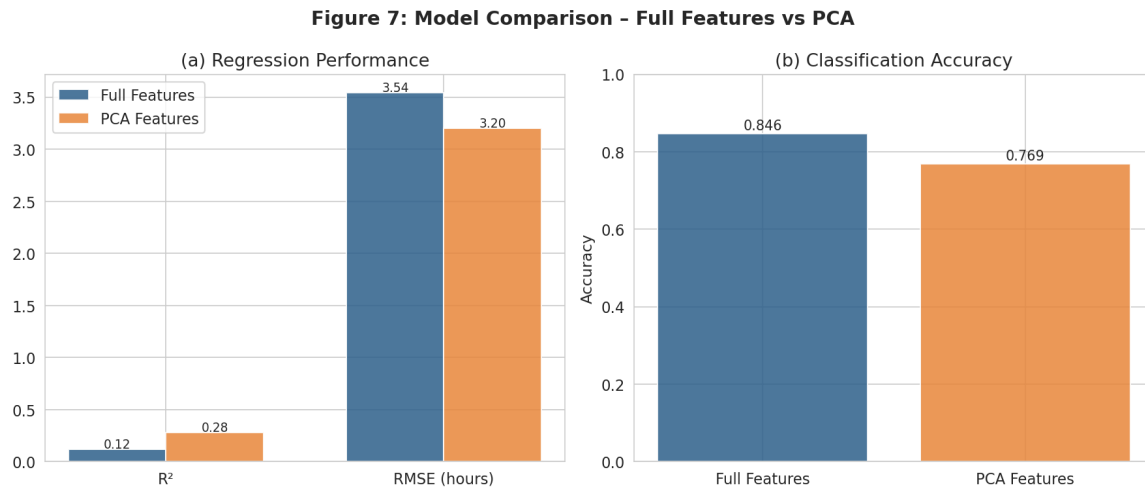


Fig. 7. Comparison of full-feature vs. PCA-feature performance for regression (R^2 and RMSE) and classification (accuracy).

The PCA features produced a higher R^2 for regression (0.282 vs. 0.120), likely due to noise reduction. However, for classification, PCA features reduced accuracy (76.9% vs. 84.6%), suggesting that some discriminative information is lost when compressing to 10 components. Retaining all 17 features is recommended for the classification task.

3. Decisions and Conclusions

The three models offer complementary value for human resource decision-making.

Regression: The Random Forest Regression model explains only about 12–30% of the variance in absenteeism hours, which limits its use for precise hour-level forecasting. The main reason is that absenteeism is influenced by unpredictable personal circumstances not captured in the data. However, it is useful for identifying which employees are trending toward higher absence loads and for aggregate workforce planning.

Classification: The classifier is the most practically useful model, achieving 84.6% accuracy and an 82% recall for high-risk employees. HR departments could use this model to proactively flag employees at risk of high absenteeism and intervene with wellness programs or schedule adjustments. The main limitation is a 25% false positive rate on the “High” class, meaning some low-risk employees would be incorrectly flagged.

Clustering: The K-Means model identifies a high-risk cluster (Cluster 6) with nearly four times the average absenteeism. This is actionable: the organization could target Cluster 6 employees with personalized health support. Other clusters reveal that most employees fall into low-absence groups that differ mainly in commute costs and age, suggesting that logistical support such as transport subsidies may reduce absenteeism for specific groups.

Overall, the dataset’s weak linear correlations and right-skewed target make precise prediction difficult. Nonetheless, the models provide actionable insights. Reason for absence consistently emerges as the most important predictor across all models, suggesting that tracking medical appointment categories is the single most valuable data point for absence management.

Future improvements could include adding longitudinal data and enriching the feature set with health indicators or job satisfaction scores.

Notes

Generative AI such as Claude, CoPilot, and Gemini are used to assisted in research, code generation, figure design, document formatting, and writing.

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